Taking powers of 2×2 matrices

We have shown that the eigenvalues of a 2×2 matrix A are

$$\frac{\mathrm{trace}(A)}{2} \pm \sqrt{\left(\frac{\mathrm{trace}(A)}{2}\right)^2 - \det(A)}$$

So A either has two distinct eigenvalues (each with algebraic multiplicity 1) or a single eigenvalue (with algebraic multiplicity 2). If A has two distinct eigenvalues, it is diagonalizable. If A has a single eigenvalue, A is diagonalizable or not depending on whether its eigenvalue has geometric multiplicity 2 or 1.

Case 1: Diagonalizable

To take the nth power of a diagonalizable matrix A, first find its diagonalization:

$$A = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}^{-1}$$

Taking the nth power of both sides yields

$$A^n = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix} \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}^{-1}$$

And after completing the two matrix multiplications on the right, we have a closed formula for A^n.

Example:

$A = \begin{pmatrix} -6 & 2 \\ -36 & 11 \end{pmatrix}$

Since A has eigenvalues 2 and 3 with corresponding eigenvectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 9 \end{pmatrix}$, we have the diagonalization

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix}^{-1}$$

So we have

$$A^n = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & 3^n \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix}^{-1}$$

$$A^n = \begin{pmatrix} -6 & 2 \\ -36 & 11 \end{pmatrix}^n = \begin{pmatrix} 9 \cdot 2^n - 8 \cdot 3^n & -2(2^n - 3^n) \\ 36(2^n - 3^n) & -8 \cdot 2^n + 9 \cdot 3^n \end{pmatrix}$$

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Case 1: Not diagonalizable

In this case the 2×2 matrix has a single eigenvalue λ. By the Schur decomposition theorem, it is similar to an upper triangular matrix, B. Any such matrix, being upper triangular, has its eigenvalues on its diagonal. Therefore we may take B to be of the form

$$B = \begin{pmatrix} \lambda & k \\ 0 & \lambda \end{pmatrix}$$

We have the following fact, which is just a computation:

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} \frac{1}{k} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \lambda & k \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$$

We may therefore take B to have the form $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$.

Since A has an eigenvector $v_1 = \begin{pmatrix} a \\ b \end{pmatrix}$, we may find a second basis vector $v_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ for our similarity equation by solving the linear system resulting from the following calculation:

$$A \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} \lambda c + a \\ \lambda d + b \end{pmatrix}$$

We now have

$$A = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}^{-1}$$

Taking the nth power of both sides yields

$$A^n = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^n \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}^{-1}$$

It remains to determine the result of $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^n$.

By the D+N decomposition theorem, we have that $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ may be decomposed uniquely into the sum of a diagonalizable matrix D and a nilpotent matrix N where D and N commute multiplicatively. In our case we have

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

The former is diagonal, the latter is nilpotent, and they commute.

Since $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^m = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ for $m \geq 2$, we have by the binomial theorem:

$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^n = \left(\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right)^n =$

$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}^n + \binom{n}{1} \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}^{n-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \binom{n}{2} \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}^{n-2} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 + \cdots + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^n =$

$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}^n + n \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}^{n-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} =$

$\begin{pmatrix} \lambda^n & n \cdot \lambda^{n-1} \\ 0 & \lambda^n \end{pmatrix}$

Note that for the second equality in this calculation, it is important that D and N commute!

After completing the two matrix multiplications on the right, we now have a closed formula for A^n.

$$A^n = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix} \begin{pmatrix} \lambda^n & n \cdot \lambda^{n-1} \\ 0 & \lambda^n \end{pmatrix} \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}^{-1}$$

Example:

$A = \begin{pmatrix} 7 & -12 \\ 3 & -5 \end{pmatrix}$

A has the single eigenvalue 1 with corresponding eigenvector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$. By the arguments above, we have that A is similar to $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. We can find the second vector for our change of basis matrix by solving

$$\begin{pmatrix} 7 & -12 \\ 3 & -5 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 1 \cdot c + 2 \\ 1 \cdot d + 1 \end{pmatrix}$$

This yields $v_2 = \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix}$.

We therefore have

$$A^n = \begin{pmatrix} 7 & -12 \\ 3 & -5 \end{pmatrix}^n = \begin{pmatrix} 2 & \frac{1}{3} \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1^n & n \cdot 1^{n-1} \\ 0 & 1^n \end{pmatrix} \begin{pmatrix} 2 & \frac{1}{3} \\ 1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 1 + 6n & -12n \\ 3n & 1 - 6n \end{pmatrix}$$